F79SP Coursework: Manufacturing Processes ($D = 4$)

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1. Introduction

In this report, we use stochastic simulation to estimate the expected manufacturing time, product quality distribution, total manufacturing cost, and overall profitability for each proposal. The analysis is based on the Markov model P, which describe the transition between 3 machines, namely Machine A (State A), B (State B), C (State C), and between 3 end products, namely excellent (State P_e), satisfactory (State P_s) and defective (State P_d) products.

The aim is to determine the most efficient and profitable production route for a product that progresses through a sequence of machines following a Markov process. Three initial proposals are being compared:

- Proposal 1 - Starts all items in Machine A.
- Proposal 2 - Starts all items in Machine B.
- Proposal 3 - Randomly assigns items to either Machine A or B, with equal probability.

Based on our analysis, we find that Proposal 2 is the most suitable production route. We also provide some further consideration of the validity of the results.

2. Model and Simulation Explanation

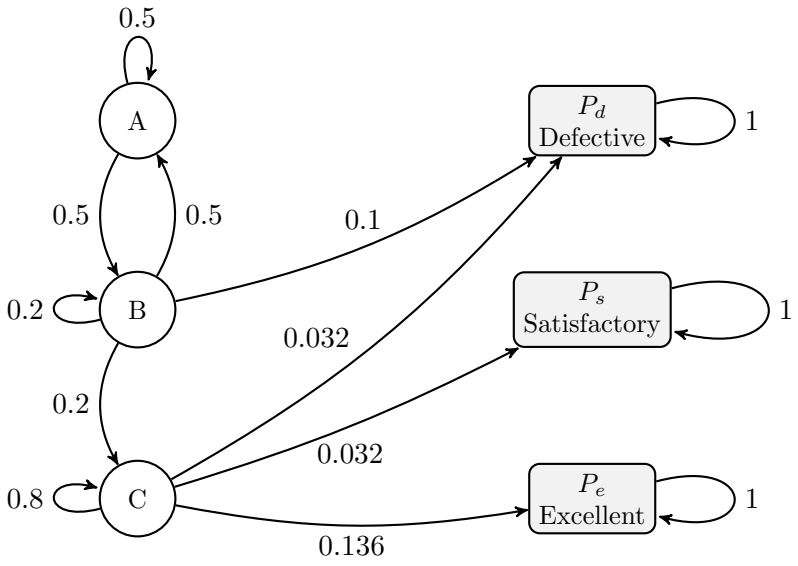


Figure 1: Transition Diagram of The Manufacturing Process' Markov Chain.

Notice that an item in Machine A has an equal chance to transition into both Machine A and B. Machine B has a broader set of possible transitions, including a 10% probability of producing a defective product directly. Machine C, on the other hand, can continue reprocessing items with a

high self-transition probability of 80%, or produce one of the three product outcomes according to probabilities $p_{3,4} = 0.136$, $p_{3,5} = 0.032$ and $p_{3,6} = 0.032$. (See Appendix A.1.). If an item reaches states P_i, where $i \in \{e, s, d\}$, then it terminates, as these are absorbing states.

At the start of each trial, a material cost of \$2 is incurred. As the product progresses, each transition represents one unit of manufacturing time. A cost of \$0.10 per unit time is charged while in Machines A and B, and \$0.20 while in Machine C. When the item reaches an absorbing state, a selling price of \$9, \$6, or \$0 is recorded for Excellent, Satisfactory, and Defective products respectively.

The simulation function `Simulate_Manufacturing_Process(n, X_1)` is implemented to process the large number of trials required for estimation. Each trial tracks total time of manufacturing process, total time spent in Machines A and B, total hit frequencies of each absorbing state, incurred costs, and obtained revenue. At the end of the simulation, the averages across all trials are taken to estimate the expected total manufacturing time, product quality proportions, total manufacturing costs and profit. We note that profit is total revenue less total manufacturing costs and material cost, and this calculation must be applied before averaging across all trials.

The 3 proposals are studied by changing the initial distribution `X_1` in the function parameters. By repeating the simulation over a large number of trials ($n = 10000$), the results approximate the theoretical expectations of these performance measures. Relevant analytical calculations will be provided in Appendix to support our justification.

3. Results

To determine the best proposal, 4 metrics have been both simulated and theoretically verified to be put into consideration. The 4 metrics being:

- Expected Manufacturing Time
- Expected Product Quality Distribution
- Expected Total Manufacturing Cost
- Expected Profit

The expected manufacturing time measures the average number of steps required for a product to reach a final quality outcome. Each "step" represents a transition between machines within the process before reaching an absorbing state (Excellent, Satisfactory, or Defective). We summarise the simulation and theoretical result in Table 1. More details of the latter can be found in Appendix A.2.

Proposal	1	2	3
Simulation result	12.0769	**10.0015**	11.0409
Theoretical result	12	**10**	11

Table 1: Comparison of Proposals 1,2 and 3 for Expected Manufacturing Time

From Table 1, proposal 2 achieves the lowest expected manufacturing time both in simulation and theory, indicating higher efficiency compared to proposals 1 and 3. The close alignment between simulated and theoretical results further validates the Markov chain model's accuracy in estimating production duration.

Next, the expected product quality distribution describes the probability that a product ends up in each of the three absorbing states. The simulated results are summarised in Table 2 below.

Proposal	1	2	3
Excellent	0.4594	0.4634	0.4603
Satisfactory	0.1041	0.0997	0.1048
Defective	0.4365	0.4369	0.4349

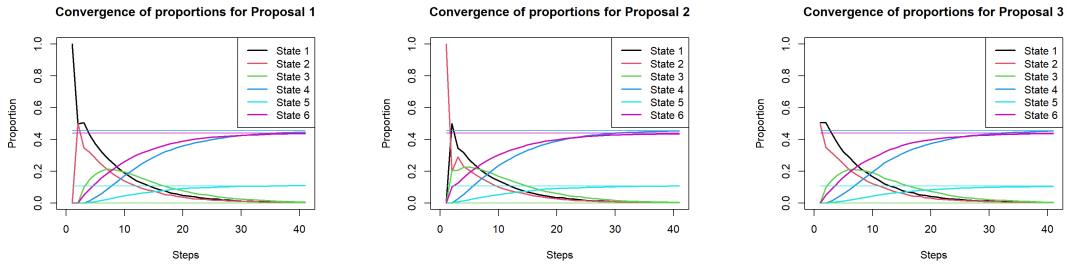
Table 2: Comparison of Proposals 1,2 and 3 for Simulated Expected Product Quality Distribution

The derivation of the theoretical values are summarised in the Table 3 below. More details on the derivation in Appendix A.3.

Proposal	1	2	3
Excellent	0.4533	0.4533	0.4533
Satisfactory	0.1067	0.1067	0.1067
Defective	0.44	0.44	0.44

Table 3: Comparison of Proposals 1,2 and 3 for Theoretical Expected Product Quality Distribution

To ensure product quality proportions eventually converge to theoretical values, we have run a simulation of 10000 trials with 40 time steps with the results shown in Figure 2 below.



(a) Proposal 1 (b) Proposal 2 (c) Proposal 3

Figure 2: Long-term proportion of product quality for 3 proposals

Note that the simulation and theoretical results in Tables 2 and 3 show consistent proportions across proposals. Approximately 45% of products are expected to be Excellent, 10% Satisfactory, and 44% Defective. As these probabilities are independent of the starting machine, all proposals yield nearly identical product quality distributions. Hence, this metric fails to conclusively distinguish the best of the 3 proposals. However, the near homogeneity between the simulated and theoretical results regulates the correctness of the findings.

Furthermore, the expected total manufacturing cost captures the average total cost incurred per product from start to completion, excluding material costs at the start of production but including machine costs accumulated along the manufacturing path. Theoretical results were computed using the expected number of visits to each transient state multiplied by their respective costs (See Appendix A.4.). The initial material cost will be considered when calculating profits later. The summary of expected total manufacturing cost is summarised in Table 4 as follows.

Proposal	1	2	3
Simulation result	1.5448	**1.3370**	1.4404
Theoretical result	1.5333	**1.3333**	1.4333

Table 4: Comparison of Proposals 1,2 and 3 for Expected Total Manufacturing Cost

Table 4 shows that proposal 2 incurs the lowest expected total manufacturing cost among the three, both theoretically and in simulation. This is primarily due to the reduced average time spent within the costly Machine A and B stages. The close agreement between theoretical and simulated results supports the robustness of the total manufacturing cost estimation.

Lastly, expected total profit is obtained by subtracting the expected manufacturing cost and the initial material cost from the expected revenue. The revenue depends on the expected probabilities of producing Excellent, Satisfactory, or Defective products, weighted by their respective prices. Summary on the results for both simulations and theoretical are detailed in Table 5 below. More details on the theoretical calculations can be found in Appendix A.5.

Proposal	1	2	3
Simulation result	1.2144	**1.4318**	1.3311
Theoretical result	1.1866	**1.3866**	1.2866

Table 5: Comparison of Proposals 1,2 and 3 for Expected Profit

As shown in Table 5, proposal 2 clearly yields the highest expected profit, confirming it as the most cost-efficient and productive manufacturing configuration. Proposal 3, which randomly starts between Machines A and B with equal probability, produces an intermediate profit between the first 2 proposals as expected. Theoretical and simulated outcomes are quite close, indicating the accuracy of the simulation model.

4. Conclusion

Overall, the consistency between theoretical and simulated values across all four metrics demonstrates that the proposed Markov model successfully captures the key dynamics of the manufacturing process. This also verifies the accuracy and enhances the strength of the argument of the practical simulation. Proposal 2 emerges as the most optimal manufacturing production configuration, balancing shorter production time, lower costs, and higher profit margins without compromising on product quality.

As a point of consideration, more trials can be conducted to see simulation results of expected total profits converge more towards the theoretical results. This was not done currently considering both the closeness of the simulation to theory, while having to balance the exponentially marginal increasing simulation run times with the diminishing marginal increase in convergence. Finally, we take note that these results can only be reproduced if a large enough amount of products have been produced due to the Law of Large Numbers. This means that the best course of action is for the company to employ proposal 2 and produce mass scale, with the assumption that there is as much demand for as much product the company can produce.

Appendix A. Outline of Calculations

A.1. Markov Chain Model.

Transition matrix of Model P as follows:

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 & 0 & 0 & 0 \\ 0.5 & 0.2 & 0.2 & 0 & 0 & 0.1 \\ 0 & 0 & 0.8 & 0.136 & 0.032 & 0.032 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Calculation of transition probabilities for Model P:
$p_{3,4} = 0.2(0.6 + \frac{4}{50}) = 0.136$. Similarly, $p_{3,5} = 0.2(0.2 - \frac{4}{100}) = 0.032$ and $p_{3,6} = 0.2(0.2 - \frac{4}{100}) = 0.032$.

A.2. Expected Manufacturing Time.

Consider Model P redefined as $P' = \begin{pmatrix} T & M \\ 0 & I \end{pmatrix}$ where $T = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0.5 & 0.2 & 0.2 \\ 0 & 0 & 0.8 \end{pmatrix}$ forms the transient matrix, $M = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0.1 \\ 0.136 & 0.032 & 0.032 \end{pmatrix}$ forms the matrix that maps transient states to absorbing states, 0 is the 3×3 zero matrix as products cannot re-enter the machines and I is the 3×3 Identity matrix as products cannot be changed of quality once produced.

Now, define Z as the expected time spent matrix in each machine starting from each machine, then, Z will be the sum of the initial state I and all possible n-th timestep variations (T^n) before being absorbed, thus,

$$\begin{aligned} Z &= I + T + T^2 + \dots \\ \implies (I - T)Z &= I + T + T^2 + \dots - T - T^2 - T^3 - \dots \\ \implies Z &= (I - T)^{-1} \\ \therefore Z &= \begin{pmatrix} 5.333\bar{3} & 3.333\bar{3} & 3.333\bar{3} \\ 3.333\bar{3} & 3.333\bar{3} & 3.333\bar{3} \\ 0 & 0 & 5 \end{pmatrix} \end{aligned}$$

Then, define t_j where $j \in \{A, B, A \text{ or } B\}$ means the manufacturing time if process starts at j. Since expected manufacturing time is the sum of all expected time spent in each machine starting from each machine until it gets absorbed, this gives $\mathbb{E}[t_A] = z_{1,1} + z_{1,2} + z_{1,3} = 12$ and $\mathbb{E}[t_B] = z_{2,1} + z_{2,2} + z_{2,3} = 10$. Due to the nature of the matrix, only the expected manufacturing times of proposal 1 and 2 can be shown. To find for proposal 3, since process starts at either Machine A or B with equal probability, consider $\mathbb{E}[t_{A \text{ or } B}] = \mathbb{E}[0.5t_A + 0.5t_B] = 0.5\mathbb{E}[t_A] + 0.5\mathbb{E}[t_B] = 11$.

A.3. Expected Proportion of Products.

Define $\alpha_i^C = \mathbb{P}(\text{Markov Chain } \{P_n\} \text{ eventually absorbed in absorbing state } C | P_0 = i)$. Then, define H as hitting probability matrix starting from each machine to eventually each product quality.

Hence, H will be the probability to transition from every possible n-th timestep (T^n), including the initial state (I), to each absorbing state. Hence, $H = ZM$, which gives,

$$H = \begin{pmatrix} 0.4533 & 0.1067 & 0.44 \\ 0.4533 & 0.1067 & 0.44 \\ 0.68 & 0.16 & 0.16 \end{pmatrix}$$

Resulting in $\alpha_A^{P_e} = h_{1,1} = 0.4533, \alpha_A^{P_s} = h_{1,2} = 0.1067, \alpha_A^{P_d} = h_{1,3} = 0.44, \alpha_B^{P_e} = h_{2,1} = 0.4533, \alpha_B^{P_s} = h_{2,2} = 0.1067$ and $\alpha_B^{P_d} = h_{2,3} = 0.44$.

We note that H matrix only shows hitting probabilities of product qualities starting from each machine. Hence, only the hitting probabilities proposal 1 and 2 can be shown. To find for proposal 3, since process starts at either Machine A or B with equal probability, consider $\alpha_{A \text{ or } B}^{P_i} = \mathbb{E}[0.5\alpha_A^{P_i} + 0.5\alpha_B^{P_i}] = 0.5\mathbb{E}[\alpha_A^{P_i}] + 0.5\mathbb{E}[\alpha_B^{P_i}]$ where $i \in \{e, s, d\}$, thus, $\alpha_{A \text{ or } B}^{P_e} = 0.4533, \alpha_{A \text{ or } B}^{P_s} = 0.1067$ and $\alpha_{A \text{ or } B}^{P_d} = 0.44$.

A.4. Expected Total Manufacturing Costs.

Define L_j as manufacturing costs if process starts at j, then the expected manufacturing costs follows as $\mathbb{E}[L_A] = 0.1z_{1,1} + 0.1z_{1,2} + 0.2z_{1,3} = 1.5333$ and $\mathbb{E}[L_B] = 0.1z_{2,1} + 0.1z_{2,2} + 0.2z_{2,3} = 1.3333$ due to excluding the initial material cost of \$2. Similarly to **A.2.**, $\mathbb{E}[L_{A \text{ or } B}] = 0.5\mathbb{E}[L_A] + 0.5\mathbb{E}[L_B] = 1.4333$

A.5. Expected Profit.

Define R_j as revenue if manufacturing process starts from j, then it follows that $\mathbb{E}[R_j] = 9\alpha_j^{P_e} + 6\alpha_j^{P_s} + 0\alpha_j^{P_d}$. We note that profit is revenue less total cost, and total cost is the sum of the total manufacturing cost and initial material cost, implying that expected profit will be expected revenue less expected total cost. Therefore, defining Pr_j as profit if manufacturing process starts from j, then $\mathbb{E}[Pr_j]$ for each j are 1.1866, 1.3866 and 1.2866 respectively.